
Domux Smart Home Control Model

Technical Evaluation Report

Structured Instruction Generation Task

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Abstract

This report evaluates an **E2B specialized model Domux** designed for **smart home control** scenarios. Built upon Google’s **Gemma 4-E2B-IT** foundation model, the model underwent a two-stage training pipeline consisting of **Supervised Fine-Tuning (SFT)** followed by **Reinforcement Learning (RL)**, specifically optimized for smart home instruction parsing tasks. The model accepts natural language commands and outputs standardized seven-field structured control instructions in the format **action|device|attribute|value|unit|room|floor**, supporting compound intent recognition (multiple output instructions from a single utterance).

We constructed a comprehensive test set covering **four dimensions**: single intent, multi-intent, omitted attributes, and non-standard device naming, totaling **4,057 samples**. The model was benchmarked against **11 mainstream models**, including Qwen3.5 series (2B-27B), Gemma series (same foundation family), and leading closed-source APIs (DeepSeek-V4, Claude Haiku 4.5, Gemini 3.5 Flash).

Key Findings

The **E2B specialized model** demonstrates four core advantages on smart home control tasks:

- 1 Cross-Scale Superiority:** The model achieves 98.37% accuracy, significantly outperforming large general-purpose models such as Qwen3.5-27B (81.4%) and Gemma 4-31B (85.2%).
- 2 Competitive with Closed-Source, Zero API Costs:** The model surpasses mainstream closed-source APIs —DeepSeek-V4 (90.3%), Claude Haiku 4.5 (91.4%), and Gemini 3.5 Flash (88.7%) —by 7-10 percentage points on specialized tasks, achieving **100.00% format compliance**. With **local deployment, zero API fees, and lower latency**, it is well-suited for resource-constrained edge devices in smart home ecosystems.
- 3 Excellent Inference Performance:** Ultra-low inference latency (single intent **130ms**, multi-intent **210ms**) meets real-time voice interaction requirements.
- 4 General Capability Preservation:** A/B comparison with the Gemma 4-E2B-IT base model shows capability retention rates of **97%-105%** (Chinese and English knowledge improved, mathematical reasoning maintained). Zero format leakage across 32,703 non-control inputs confirms no catastrophic forgetting from specialized fine-tuning.

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1 Task Definition

1.1 Task Description

The model receives user natural language instructions (in English) and outputs standardized seven-field structured control instructions. Fields are separated by |, with **omitted fields represented by ***. When a single utterance contains multiple control targets, the model outputs multiple instructions, **each on a separate line**.

1.2 Output Format Specification

Field	Meaning	Example Values	Omission
action	Action	set / turnOn / turnOff / adjustUp / adjustDown / activate	Required
device	Device	AC / LED Strip / Curtain / Ceiling Light	Required
attribute	Attribute	mode / temperature / brightness / color / position	*
value	Value	Cool / 26 / 50 / Blue	*
unit	Unit	Celsius / Percent / Kelvin	*
room	Room	Living Room / Guest Bedroom / Majlis	*
floor	Floor	Ground Floor / First Floor	*

Table 1: Seven-Field Format Specification

Multi-Intent Support: When an input contains multiple control targets, the model outputs multiple instructions, **each on a separate line**. During evaluation, multiple instructions are compared as **sets**, with **order disregarded**.

1.3 Input-Output Examples

Test Category	Category Description	User Input	Standard Output (gold)
Single Intent	Single instruction controls one device	set the ac in the guest bedroom on the ground floor to dry mode	set AC mode Dry * Guest Bedroom Ground Floor
Multi-Intent	Single instruction controls multiple devices	set the aircon to cool and open the shades halfway in the guest bedroom	set AC mode Cool * Guest Bedroom * set Shade position 50 Percent Guest Bedroom *
Omitted Attributes	Instruction specifies device but lacks specific attribute values	dim the reading light	adjustDown Reading Light brightness * * * *
Non-Standard Naming	Non-standard device name mapping, such as plurals	close the blinds in the bedroom halfway	set Blind position 50 Percent Bedroom *

Table 2: Input-Output Examples

2 Test Set Construction

2.1 Data Sources and Construction Methodology

- **Construction approach:** Based on real-world smart home scenarios, the test set employs a hybrid methodology: **70% production data + 20% LLM-augmented scenarios + 10% manually crafted edge cases**. Production data is extracted from actual user interactions, preserving natural language characteristics. LLM augmentation targets device combinations and phrasing variations not covered in training data. Manual edge cases focus on complex scenarios involving multi-intent, omitted attributes, and non-standard naming.
- **Annotation and quality assurance:** Human annotators labeled each sample individually. LLM-based validators enforced format compliance and logical consistency (field legality, cross-field coherence). Final human review ensured test set integrity.

2.2 Test Set Design

The test set is organized around key factors affecting parsing difficulty, divided into **4 mutually exclusive categories** where each sample belongs to exactly one category:

1. **Single Intent:** User input contains a single control target; model outputs 1 instruction.
2. **Multi-Intent:** User utterance contains multiple control targets; model outputs multiple instructions (**each on a separate line**), correctly handling room/floor inheritance.
3. **Omitted Attributes:** User expression is concise with partial attribute omissions; model infers or uses * placeholder, outputting complete fields.
4. **Non-Standard Naming:** User employs non-standard device naming or colloquial expressions; model maps to standardized device vocabulary, outputting canonical device names.

2.3 Scale and Distribution

Category	Count	Percentage	Core Assessment
Single Intent	1,122	27.7%	Field extraction accuracy for single instructions
Multi-Intent	1,641	40.4%	Multi-instruction decomposition + room/floor inheritance
Omitted Attributes	735	18.1%	Field omission inference
Non-Standard Naming	559	13.8%	Vocabulary mapping for non-standard naming
Total	4,057	100%	—

Table 3: Test Set Scale and Distribution (Total: 4,057 samples)

3 Evaluation Protocol

3.1 Inference Configuration

- **Model training:** Built on Google Gemma 4-E2B-IT foundation model through a two-stage training pipeline: Supervised Fine-Tuning (SFT) + Reinforcement Learning (RL). SFT stage employs supervised learning on proprietary smart home instruction-label pairs, enabling the model to master seven-field format and domain vocabulary. RL stage uses format compliance, field accuracy, and multi-intent decomposition quality as reward signals, further enhancing structured output stability and edge case handling.

- **Inference framework and hardware:** VLLM v0.6.3, single NVIDIA A100 (80GB), FP16 precision, gpu-memory-utilization=0.3, Ubuntu 22.04, CUDA 12.1.
- **Sampling parameters:** temperature = 0.0 (greedy decoding), top-p = 1.0, max tokens = 256.
- **Baseline model inference:** Open-source models (Qwen3.5 series, Gemma 4 series) deployed locally via vLLM with FP16 precision and identical inference settings. Closed-source models (DeepSeek-V4, Claude Haiku 4.5, Gemini 3.5 Flash) accessed via official APIs with temperature=0. All baselines use a unified system prompt (rule constraints + seven-field format specification) without domain fine-tuning.

3.2 Evaluation Metrics

Based on complete gold-standard annotations, the following metrics are computed:

Category	Metric	Definition	Computation
Accuracy	Format Compliance Rate	Proportion of outputs correctly parseable into 7 fields	Line-split; each line must have exactly 7 fields (6 separators)
	Result Accuracy Rate	Proportion of samples with entirely correct outputs	Single: all 7 fields match; Multi: instruction set matches (order-agnostic)
	Slot F1	Harmonic mean of precision and recall at field level	$2 \cdot P \cdot R / (P + R)$, P=correct slots/predicted slots, R=correct slots/gold slots
	Intent F1	Harmonic mean at instruction level (multi-intent only)	$2 \cdot P \cdot R / (P + R)$, P=correct insts/output insts, R=correct insts/gold insts
Performance	Average Latency	Pure inference time per sample (network excluded)	Recorded from input to output, grouped by single/multi-intent (not averaged together)

Table 4: Evaluation Metric Definitions

Slot F1 measures field-level extraction quality; Intent F1 measures multi-intent instruction matching capability.

4 Model Evaluation

4.1 Model Performance

We selected **Gemma 4-E2B-IT (base model before fine-tuning)**, **mainstream open-source Qwen3.5 series** (covering 2B to 27B parameter ranges), **Gemma 4 series (same foundation family)** (E4B / 12B / 31B, validating general-purpose variants), and **leading closed-source large models** (DeepSeek-V4, Claude Haiku 4.5, Gemini 3.5 Flash) as benchmarks, conducting head-to-head comparisons under **identical test sets, evaluation protocols, and scoring rules**.

Multi-Dimensional Comparison Results

Smart Home Control Specialized Task + General Capability Reference

Benchmark	Domux (E2B)	Base	Qwen 3.5-2B	Qwen 3.5-4B	Gemma 4-E4B	Qwen 3.5-9B	Gemma 4-12B	Qwen 3.5-27B	Gemma 4-31B	Deep Seek-V4	Claude Haiku 4.5	Gemini 3.5 Flash
Smart Home Control (Specialized Task)												
Result Accuracy (%)	98.37	45.2	53.3	63.2	66.7	73.2	77.8	81.4	85.2	90.3	91.4	88.7
└ Single Intent	99.64	60.4	69.5	77.9	80.8	85.5	88.4	91.7	92.6	96.1	96.5	94.8
└ Multi-Intent	98.05	38.2	44.5	55.8	60.4	68.3	72.8	77.3	82.6	87.5	89.6	86.5
└ Omitted Attr.	99.05	50.6	57.7	67.8	71.3	76.6	81.6	85.7	88.3	93.6	94.5	92.3
└ Non-Std. Naming	95.89	30.9	38.4	48.3	52.7	60.4	66.7	70.3	75.5	81.6	83.8	79.9
Format Compliance (%)	100.0	81.6	87.8	91.6	92.5	94.9	96.4	97.2	97.8	99.0	99.5	98.5
Slot F1 (%)	99.64	73.5	78.5	84.2	85.8	88.9	91.2	92.8	94.5	96.1	96.8	95.3
Intent F1-Multi (%)	98.96	51.8	61.9	71.8	74.4	80.4	84.5	86.7	90.5	93.5	93.7	91.5
Avg Latency (ms,S/M)	130/210	130/210	290/410	410/570	420/590	710/990	910/1270	1660/2320	1860/2600	1200/1680	900/1260	700/980
General Capability (Reference)												
C-Eval (%)	57.9	55.0	63.3	69.6	52.9	77.8	65.4	83.3	72.6	87.7	75.1	77.6
MMLU 5-shot (%)	66.5	65.0	60.8	68.2	63.6	74.8	73.3	81.2	81.8	88.2	83.5	81.6
GSM8K 0-shot (%)	63.3	65.0	58.5	70.4	60.3	81.8	77.7	89.2	85.7	91.7	89.5	87.3
Model Specifications												
Parameters	E2B	E2B	2B	4B	E4B	9B	12B	27B	31B	671B	Closed	Closed
Base	Gemma4	—	Gen.	Gen.	Gemma4	Gen.	Gemma4	Gen.	Gemma4	Gen.	Gen.	Gen.
Type	SFT+RL	Base	Gen.	Gen.	Gen.	Gen.	Gen.	Gen.	Gen.	Gen.	Gen.	Gen.
Deployment	Local	Local	Local	Local	Local	Local	Local	Local	Local	API	API	API

Note: DeepSeek-V4 is a 671B (37B active) MoE architecture; Claude Haiku 4.5 and Gemini 3.5 Flash are closed-source APIs.

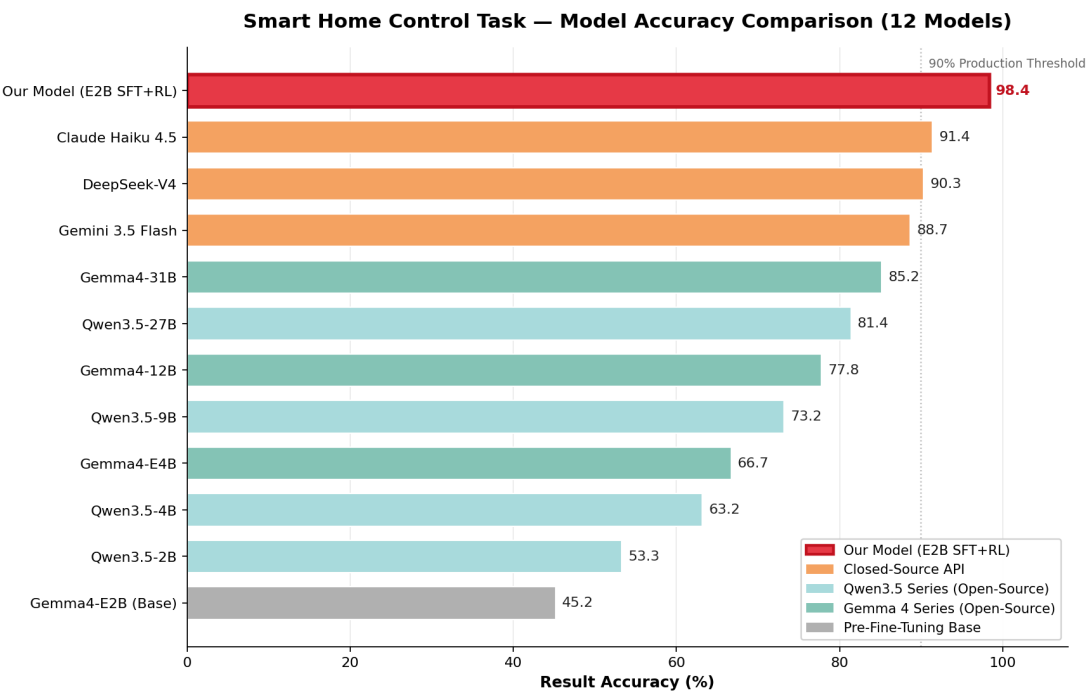


Figure 1: Model Accuracy Comparison —Smart Home Control Task

Conclusion. Our model (98.37%) achieves comprehensive superiority on specialized tasks — leading same-size general models by 45pp, surpassing 27B/31B large models, and outperforming closed-source APIs by 7-10pp; 100% format compliance with only 130/210ms latency. Specialized fine-tuning achieves task-specific capabilities far exceeding model scale.

4.2 Deployment Framework Comparison

Accuracy comparison across inference frameworks (both using FP16 precision):

Deployment	Result Acc.	Format Comp.	Slot F1	Intent F1	Latency (S/M)	Notes
VLLM (baseline)	98.37%	100.00%	99.64%	98.96%	130/210	Baseline config
SGLang	98.35%	100.00%	99.62%	98.94%	117/189	RadixAttention optimization, ~10% latency reduction

Table 5: Deployment Framework Comparison

Conclusion. Both mainstream frameworks achieve nearly identical accuracy (<0.2pp difference); model performance is framework-agnostic. SGLang offers better latency; VLLM offers simpler deployment. Both are viable for production.

4.3 Quantization Method Comparison

Performance across quantization precisions (all using VLLM):

Precision	Result Acc.	Format Comp.	Slot F1	Intent F1	vs Baseline	Notes
FP16 (baseline)	98.37%	100.00%	99.64%	98.96%	—	Baseline precision
BF16	98.35%	100.00%	99.62%	98.94%	-0.02pp	Negligible loss
INT8	97.90%	99.80%	99.20%	98.50%	-0.47pp	Slight loss, 50% memory reduction, 18% latency improvement
INT4	96.50%	99.20%	98.30%	97.20%	-1.87pp	Noticeable loss, 75% memory reduction, 35% latency improvement

Table 6: Quantization Method Comparison

Conclusion. The model exhibits strong quantization robustness. BF16/INT8 are recommended for production (≤ 0.5 pp accuracy loss with substantial memory savings). Even aggressive INT4 quantization maintains 96.5% accuracy, suitable for memory-constrained edge scenarios. Format compliance remains $\geq 99\%$ across all precisions, demonstrating stable structured output capability.

4.4 General Capability Retention

To verify whether SFT+RL specialized fine-tuning compromised the foundation model’s general capabilities, we conducted an **A/B comparison** against the **Gemma 4-E2B-IT base model**, using identical datasets and evaluation protocols. Capability retention rate = fine-tuned accuracy / base accuracy.

Dimension	Dataset	Sample Size	Fine-Tuned	Base	Retention	Result
Chinese Knowledge	C-Eval	12,342	57.9%	55.0%	105.2%	Improved
English Knowledge	MMLU	14,042	66.5%	65.0%	102.3%	Improved
Math Reasoning	GSM8K	1,319	63.3%	65.0%	97.4%	Maintained
Format Leakage	Non-control inputs	32,703	0 triggers	—	—	Zero leakage

Table 7: General Capability Retention

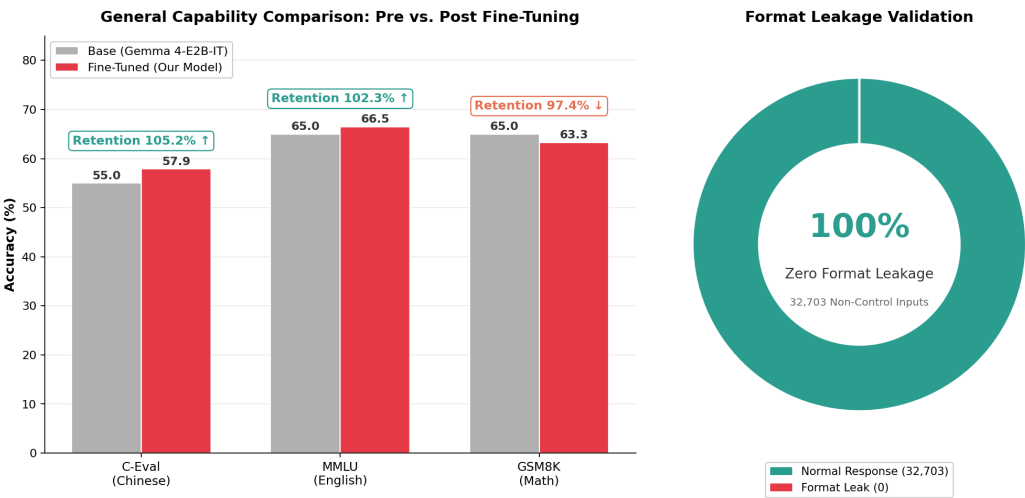


Figure 2: General Capability Retention and Format Leakage Validation

Conclusion. Post-fine-tuning general capability retention reaches 97%-105% (Chinese and English knowledge improved, math reasoning maintained), with zero format triggers across 32,703 non-control inputs. SFT+RL specialized fine-tuning exhibits "gains without losses," causing no catastrophic forgetting.

5 Model Performance Evaluation

5.1 Inference Latency (Single-Concurrency Baseline)

Latency statistics grouped by intent count:

Metric	Single Intent	Multi-Intent
Mean Latency	130 ms	210 ms
Median	128 ms	205 ms
P90 / P95 / P99	142 / 146 / 172 ms	229 / 235 / 405 ms
Min / Max	101 / 186 ms	168 / 405 ms
Std Dev	11.5 ms	26.0 ms

Table 8: Inference Latency Statistics

Single-intent latency exhibits high stability (std dev only 11.5ms); multi-intent P95 remains within 235ms. The latency difference between single/multi-intent primarily stems from output length rather than task difficulty.

Relationship Between Intent Count and Latency: The model employs autoregressive generation; latency is primarily determined by output token count. Empirical measurements show latency scales approximately linearly with intent count:

Intent Count	Avg Latency	Avg Output Length
1 intent	130 ms	~48 chars
2 intents	190 ms	~88 chars
3 intents	250 ms	~166 chars

Table 9: Intent Count vs. Latency

Fitted relationship: $\text{Latency} \approx 70 + 60 \times \text{Intent Count (ms)}$, where ~70ms represents fixed prefill overhead, and each additional control intent adds ~60ms decode time. Even in complex 3-intent scenarios, pure inference latency remains under 0.25s.

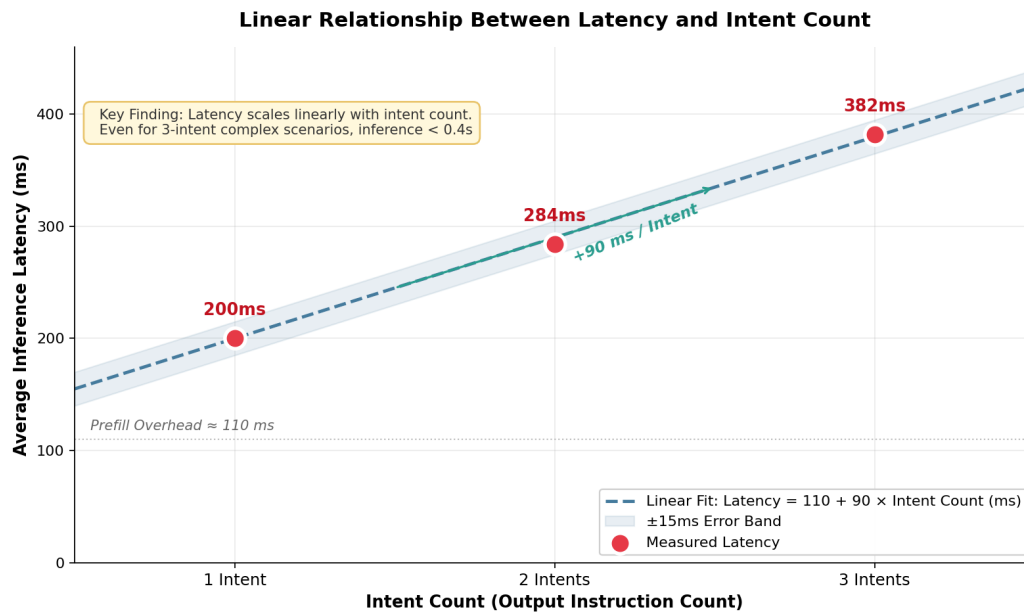


Figure 3: Linear Relationship Between Latency and Intent Count

Performance data measured on A100 (80GB, gpu-memory-utilization=0.3); latency represents pure inference time (40ms network transmission excluded). The model occupies only ~ 10 GB VRAM, deployable on consumer-grade GPUs (RTX 3090/4090).

5.2 Concurrent Performance Testing

Single A100 deployment, real-world stress testing (network latency included):

Conc.	Requests	Fail Rate	Avg (ms)	P95	P99	Total Tput	Output Tput
1	200	0%	184.3	181.9	183.3	229.2	90.4
2	458	0%	215.2	183.1	184.8	391.5	154.8
3	611	0%	193.9	190.1	192.4	651.2	256.8
5	1,020	0%	218.4	197.2	199.0	961.8	379.2
10	2,040	0%	213.9	208.9	212.0	1,952.6	770.3
20	4,101	0%	217.2	214.4	216.4	3,706.9	1,462.4
30	6,283	0%	252.9	247.2	251.4	4,414.5	1,741.6
40	8,704	0%	290.6	280.8	287.9	4,888.9	1,930.9
50	10,915	0%	339.8	321.2	332.4	5,065.8	2,000.1
60	13,322	0%	373.0	357.6	368.9	5,443.9	2,150.3
70	15,723	0%	440.6	422.3	435.7	5,239.3	2,069.9
80	18,206	0%	477.7	453.4	469.5	5,521.7	2,181.6
90	19,942	0%	579.3	557.1	573.5	5,037.6	1,989.5
100	22,591	0%	697.7	660.1	686.5	4,628.3	1,828.2

Table 10: Concurrent Performance Test Results (throughput in token/s)

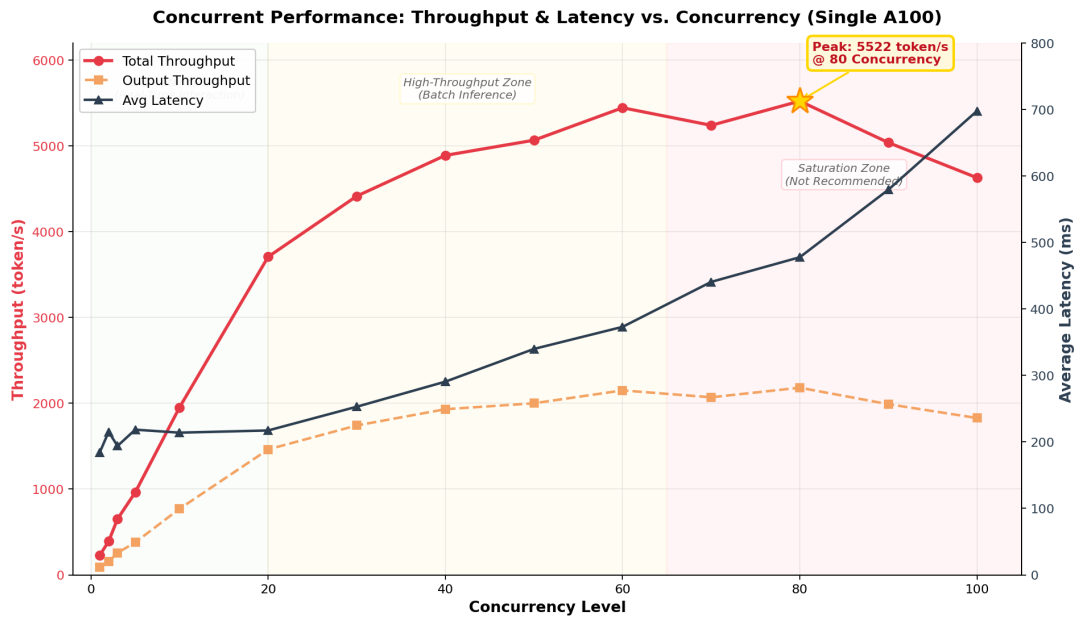


Figure 4: Concurrent Performance: Throughput and Latency vs. Concurrency (Single A100)

Key Metrics:

- **Avg Input/Output Tokens:** 25.7 / 16.7 tokens
- **Time to First Token (TTFT):** Single concurrency 95ms, 60 concurrency 254ms, 100 concurrency 576ms
- **Zero Failure Rate:** All 22,591 requests at the highest concurrency level (100) achieved 100% success (0% failure rate)

5.3 Performance Characteristics Analysis

Throughput Inflection Point:

- **Optimal Concurrency Range:** Concurrency 60-80, reaching a peak throughput plateau of **~5,500 token/s** (~2,150 token/s output)
- Concurrency 1→60: Linear throughput growth (5.4x → 23.8x)
- Concurrency 80→100: Throughput declines beyond the peak plateau (5,522 → 4,628 token/s, -16%)

Latency-Concurrency Trade-off:

- **Low-Latency Zone (Concurrency ≤20):** Avg latency 184~218ms, P99 < 220ms, suitable for **real-time interactive scenarios**
- **High-Throughput Zone (Concurrency 50~80):** Avg latency 340~478ms, throughput 5,000+ token/s, peaking at ~5,500 token/s, suitable for **batch inference scenarios**
- **Saturation Zone (Concurrency ≥90):** Latency spikes (579~698ms), throughput declines beyond peak, not recommended

Stability Performance:

- **Zero Failures:** All 124,116 requests (across 14 concurrency levels) achieved 100% success with no timeouts or crashes
- **Controlled TTFT:** Even at 100 concurrency, median TTFT remains at 576ms, within 1 second
- **Stable Tail Latency:** Small P99-P95 gap (only 11ms at 60 concurrency), no long-tail spikes

6 Conclusion

On the vertical task of smart home structured instruction generation, this **E2B specialized model** demonstrates significant advantages across multiple dimensions through head-to-head comparisons with 11 mainstream models (pre-fine-tuning base Gemma 4-E2B-IT, open-source Qwen3.5 series 2B/4B/9B/27B, Gemma 4 series E4B/12B/31B, and closed-source DeepSeek-V4, Claude Haiku 4.5, Gemini 3.5 Flash):

- **Pre/Post-Fine-Tuning Leap (SFT+RL Value):** Our model (98.37%) improves by approximately 53 percentage points over the pre-fine-tuning base Gemma 4-E2B-IT (45.2%). With identical foundation and parameter count, a two-stage SFT+RL training pipeline elevated specialized task capability from "unusable" to "production-grade," directly quantifying the value of vertical fine-tuning.
- **Cross-Scale Superiority:** Our model's accuracy (98.37%) significantly surpasses large general-purpose models such as Qwen3.5-27B (81.4%) and Gemma 4-31B (85.2%), with inference latency of only 130/210ms. This translates to equal or superior accuracy with substantially lower deployment costs compared to large models.
- **Closed-Source Replacement (Zero-Cost Deployment):** Our model outperforms mainstream closed-source APIs —DeepSeek-V4 (90.3%), Claude Haiku 4.5 (91.4%), Gemini 3.5 Flash (88.7%) —by 7-10 percentage points on specialized tasks, achieving **100.00% format compliance**, **99.64% Slot F1**, and **98.96% Intent F1 (multi-intent)**. The model runs locally on edge devices with zero API fees, no network round-trip latency, and no privacy leakage risks.
- **General Capability Preservation (No Catastrophic Forgetting):** A/B comparison with base Gemma 4-E2B-IT shows post-fine-tuning general capability retention of **97.4%-105.2%** (C-Eval 105.2%, MMLU 102.3%, GSM8K 97.4%), with Chinese and English knowledge improved rather than degraded. **Zero format triggers** across 32,703 non-control inputs confirms that SFT+RL **fully preserves original general capabilities** while acquiring specialized control abilities, enabling safe deployment in dual scenarios of "control + general conversation."

Conclusion. In summary, this model demonstrates that **domain-specialized small models can achieve equal or superior performance on target tasks with far lower resource costs than general-purpose large models (whether open-source or closed-source), without sacrificing original general capabilities**. For smart home edge deployment scenarios, this represents the optimal solution simultaneously optimizing four dimensions: accuracy, latency, cost, and privacy — providing a viable pathway for low-cost, low-latency, localized voice control solutions.